OKELLO ELLY

SOFTWARE ENGINEER | AI ENGINEERING | FULL-STACK & MOBILE DEVELOPMENT

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PROFESSIONAL SUMMARY

Software Engineering undergraduate and developer building mobile applications, web platforms, backend services, and AI-enabled products. Experience spans Kotlin-based Android development, TypeScript web applications, frontend engineering, API development, and systems prototyping. Founder of technology initiatives focused on energy optimization, sustainability, and digital innovation. Combines practical software development with systems thinking, product design, and innovation leadership.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, TypeScript, Java, Kotlin, SQL, C#, C, C++
- **Frontend & Mobile:** React, React Native, Expo, Next.js, HTML, CSS, Android Studio, Material Design
- **Backend & APIs:** FastAPI, Flask, REST APIs, Pydantic, API integration, JWT authentication, role-based access control
- **Databases:** MySQL, SQLite, relational database design, PL/SQL
- **AI & Automation:** Claude API, generative AI application development, LangGraph, CrewAI, Zapier, n8n, Lovable
- **Optimization & Reporting:** PuLP, CBC, linear programming, ReportLab, data-driven reporting
- **Tools & Engineering:** Git, GitHub, VS Code, object-oriented programming, software testing, system design
- **Networking & Security:** TCP/IP, VLANs, routing, switching, IP addressing, cybersecurity fundamentals

PROFESSIONAL EXPERIENCE

AI & Mobile Development Intern | Momentum Labs

August 2026 – Present

- Contribute to React Native mobile UI development and application feature implementation within an ongoing software project.
- Refine mobile interfaces and support iterative feature development in collaboration with the team.

Front-End AI Engineering Intern | FlyRank AI

July 2026 | One-month internship

- Completed AI and frontend engineering masterclasses, applying concepts through weekly development assignments.
- Built a final capstone project, consolidating learning through practical implementation and iterative refinement.

PROJECTS

VumaGrid | Founder & Systems Architect

2026 – Present | Prototype / Partially Implemented
GitHub: https://github.com/Elly739/VumaGrid

- Architected and built a prototype distributed energy management platform using FastAPI and Next.js, integrating simulated multi-vendor energy telemetry, battery state-of-charge modelling, asset monitoring, and an operations dashboard.
- Developed a PuLP/CBC linear-programming engine for energy-cost optimization against Kenyan time-of-use tariffs, alongside battery-capacity scenario analysis estimating annual savings, payback periods, and CO_2 reductions.
- Implemented asset anomaly checks, webhook alerts, tenant-aware API filtering, carbon analytics, PDF-based ESG reporting, and dashboard features for asset filtering, investment simulation, PDF export, and property onboarding.

Technologies: Python, FastAPI, Pydantic, PuLP, CBC, Next.js 14, TypeScript, Tailwind CSS, ReportLab, HTTP/JSON, webhooks.

SmartWaste | Founder & Software Developer

2026 – Present

- Founded a sustainability-focused initiative exploring digital tools for waste-management monitoring, collection tracking, recycling participation, and environmental reporting.

BIT System | Backend Developer & System Designer

- Designed a campus-oriented digital platform concept supporting student services and administrative workflows, working on backend functionality, relational data structures, and REST APIs using Flask, MySQL, and SQLite.

Nairobi Hidden Gems | Android & Web Development

- Developing an Android application using Kotlin and a TypeScript-based web application for discovering Nairobi locations and experiences.

CampusHub | Android & Web Development

- Built an Android student-portal application using Kotlin and Material Design, alongside development of a TypeScript-based web application for campus services.

Pioneer Africa | Digital Product Initiative

- Developing a digital initiative connecting African students and young innovators with learning pathways, mentorship, community, and opportunities.

EDUCATION

Bachelor of Science in Software Engineering

Zetech University | January 2025 – November 2027 (Expected)

CERTIFICATIONS & TECHNICAL TRAINING

- Building with the Claude API — Anthropic, 2026
- AI Fluency for Builders — Anthropic, 2026
- Teaching the AI Fluency Framework — Anthropic, 2026
- Claude 101 — Anthropic, 2026
- AI Fluency: Framework and Foundations — Anthropic, 2026
- AI Safari — Power Learn Project, 2026
- Introduction to Modern AI — Cisco Networking Academy, 2026
- Cisco Networking Academy Learn-A-Thon 2026 — Cisco, issued April 2026
- Networking Academy Learn-A-Thon 2025 — Cisco, issued January 2026
- Cybersecurity Essentials — Cisco Networking Academy, 2025
- Junior Cybersecurity Analyst Career Path — Cisco Networking Academy, 2025
- Computer Networks — Huawei, 2025
- Data Science and Analytics — HP LIFE, 2025

LEADERSHIP & COMMUNITY ENGAGEMENT

Alumni Channel Lead — Climate/Sustainability and Education | Aspire Institute Volunteer

2026 October – Present

- Support volunteer engagement and community activities across sustainability and education channels.

Community Volunteer Moderator | Aspire Leaders Program

2026

- Supported participant engagement, peer learning, and community interaction across two program cohorts.

Secretary | Zetech University Innovation Club

- Supported student innovation activities, communication, event coordination, and student engagement.
- Contributed to club initiatives focused on research, innovation, and student projects.

PROFESSIONAL DEVELOPMENT

- McKinsey Forward Program

- Global Mentorship Initiative — Career Mentorship Program
- UNITAR training in anticipatory governance, systems thinking, and strategic foresight